

Accelerating metabolic engineering through multimodal phenotype prediction with TorchCell and the Cell Graph Transformer

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Abstract

Deep learning has advanced protein structure prediction, yet metabolic engineering lags because phenotype data are scattered across heterogeneous studies lacking a shared ontology. We present TorchCell, an open-source framework that annotates literature and public datasets with a shared ontology and ingests them into a Neo4j knowledge graph, then builds deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell combining genome sequence, gene networks, metabolism, and environment. We develop the Cell Graph Transformer (CGT), a virtual-cell architecture in which biological interaction graphs constrain multi-head attention and an equivariant perturbation operator feeds multitask decoders. In *Saccharomyces cerevisiae*, CGT predicts trigenic gene-gene interactions (Pearson $r = \mathbf{0.454}$), outperforming DANGO, DCell, and Yeast9 flux balance analysis, and generalizes to knockout expression ($r = \mathbf{0.543}$) and morphology ($r = \mathbf{0.619}$). CGT recommends gene deletions for beta-carotene and betaxanthin production, pairing with autonomous biofoundries for AI-guided strain engineering.

Keywords: metabolic engineering, foundation models, gene interactions, virtual cell, knowledge graph, strain design

1 Introduction

[FILLER – replace with Intro prose.] Deep learning transformed protein structure and function prediction, yet systems biology and metabolic engineering lag because phenotype data are scattered across small, heterogeneous studies with no shared schema [1]. Prior approaches fall short: mechanistic flux balance analysis misses epistasis, and recent perturbation transformers do not consistently beat linear baselines [2]. Our own

classical baselines predict knockout fitness but fail on gene interactions (Suppl. Fig. S1), motivating a deep, biologically grounded model. Here we present TorchCell and the Cell Graph Transformer (CGT), whose data model and architecture we outline in Fig. 1.

2 Results

A shared ontology and knowledge graph unify metabolic-engineering data.

[*FILLER – R1.*] TorchCell annotates literature and public datasets with a shared experiment ontology and ingests them into a Neo4j knowledge graph; dataloaders emit deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell (Fig. 1; data scale in Suppl. Fig. S2; ontology/KG spec in Suppl. Fig. S4).

The Cell Graph Transformer: a graph-constrained virtual cell. [*FILLER – R2.*] Biological interaction graphs constrain multi-head attention through a loss aligning attention to network priors; an equivariant perturbation operator maps the reference embedding to the perturbed strain state (Fig. 2; ablations in Suppl. Fig. S3).

CGT predicts trigenic gene-gene interactions at state of the art. [*FILLER – R3.*] Classical ML fails on this task (Suppl. Fig. S1); CGT reaches Pearson $r = 0.454$, outperforming DANGO, DCell, and Yeast9 FBA (Fig. 3).

One embedding, many phenotypes: expression and morphology. [*FILLER – R4.*] The same architecture predicts knockout expression ($r = 0.543$) and single-knockout morphology ($r = 0.619$) (Fig. 4; expression diagnostics in Suppl. Fig. S5).

CGT-guided strain design with an autonomous foundry. [*FILLER – R5.*] We recommend gene deletions for beta-carotene and betaxanthin production and pair with the UIUC iBioFoundry for iterative design (Fig. 5; inference at scale in Suppl. Fig. S6).

3 Discussion

[*FILLER – Discussion.*] A strain-aware, biologically grounded representation is the differentiator; limitations include single-organism scope, morphology-data maturity, and error-bar provenance.

4 Methods

[*FILLER – Methods.*] Ontology and knowledge-graph construction; dataset and perturbation-operator formalism; CGT architecture and the graph-regularized attention loss; training, splits, metrics; baselines (DANGO, DCell, Yeast9 FBA);

inference at scale; statistics and error-bar definitions.

Supplementary information.

Acknowledgments.

Declarations

References

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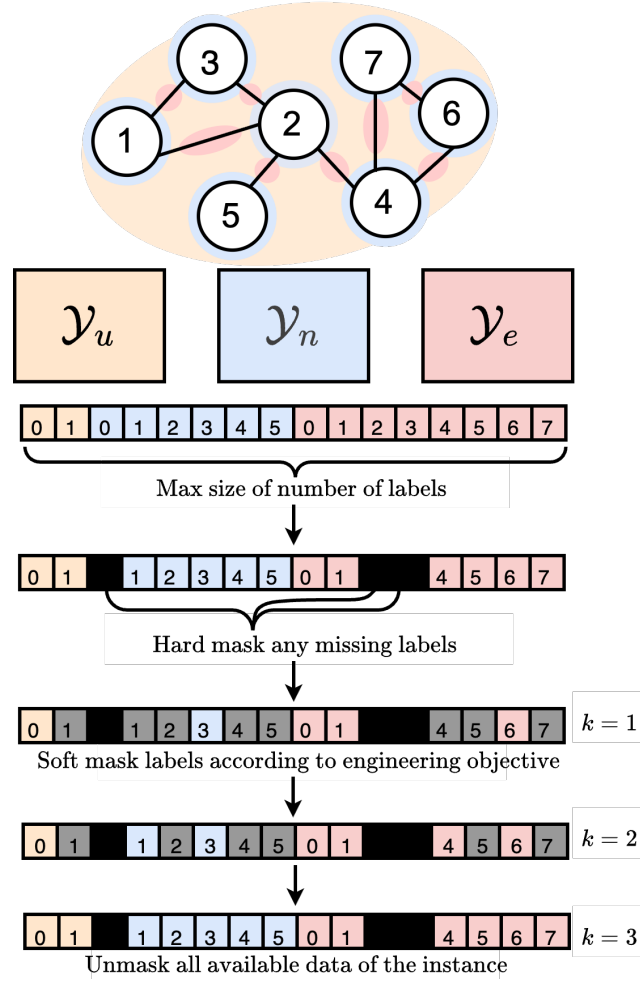


Fig. 1 TorchCell: ontology-unified knowledge graph and the perturbation-operator cell representation. Panels (a)–(d).

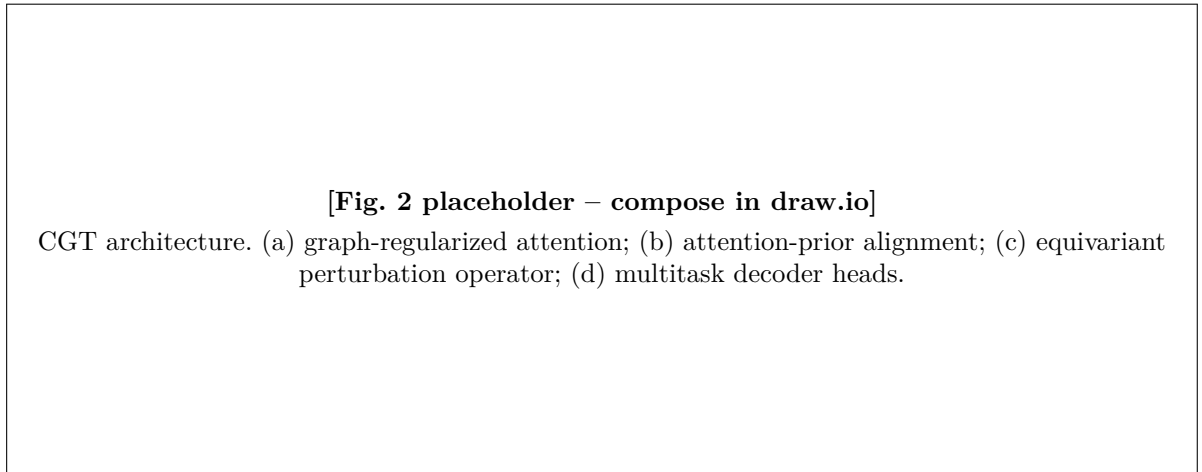


Fig. 2 The Cell Graph Transformer architecture. (a)–(d).

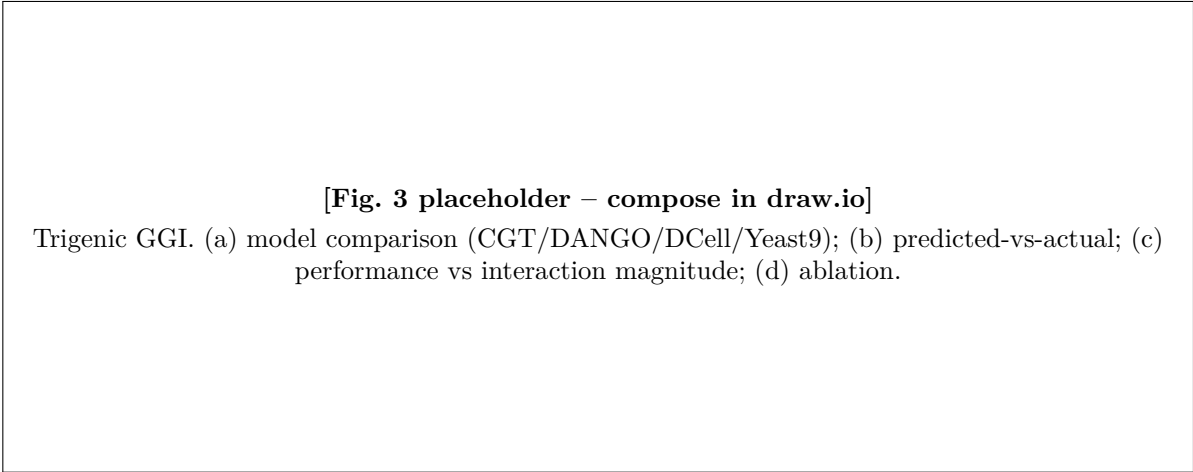


Fig. 3 State-of-the-art trigenic gene-gene interaction prediction. (a)–(d).

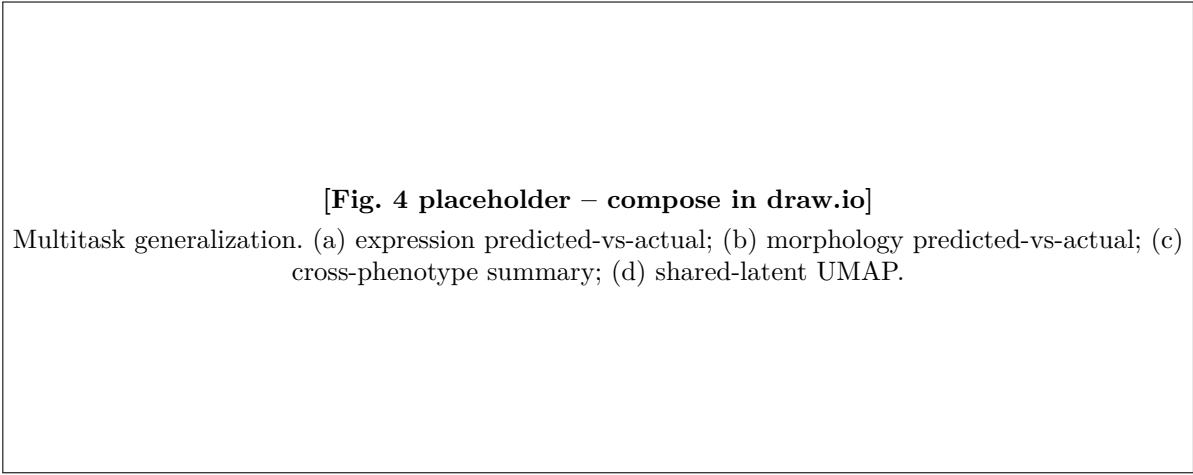


Fig. 4 One latent cell embedding generalizes across phenotype classes. (a)–(d).

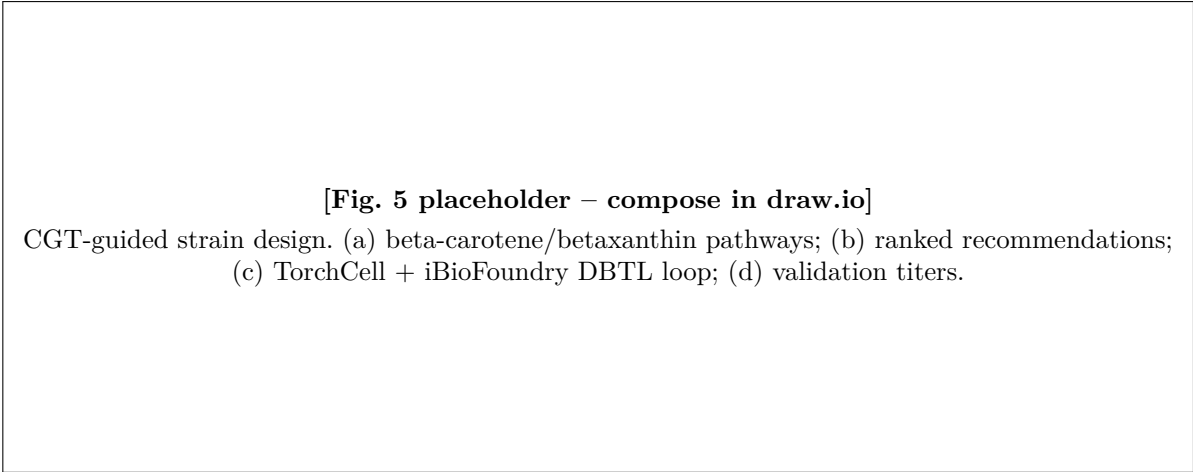


Fig. 5 CGT-guided strain design closes the DBTL loop with an autonomous foundry. (a)–(d).